



SmartEPT Central

Licensing, Billing & Client Platform — Complete Handbook

What it has, how it works, and why it is built the way it is — a single reference for the Super-Admin console, the client self-service portal, the money engine and the licensing core.

Product	SmartEPT Central — the Ametecs licensing & billing platform behind SmartEPT
Stack	Laravel 13 · PHP 8.3 · MySQL 8 · Razorpay + Stripe · RSA-2048 offline licences
Audience	Ametecs Super-Admin / Sales / Support staff, and the technical owner
Version	Handbook v1.0 · 27 July 2026
Prepared by	Ametecs India Private Limited · GST 36AAHCT0971F1ZB

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1 · What SmartEPT Central is

SmartEPT Central is the business-side platform that sits behind the SmartEPT product. Where the SmartEPT server and desktop agent do the productivity tracking for a customer, Central is where Ametecs runs the commercial side of that relationship: it issues and renews the licences that unlock the product, prices and quotes deals, collects payment through Indian and international gateways, cuts GST-compliant tax invoices, and gives every customer a self-service portal to buy, renew, download installers and raise support tickets.

In one sentence: Central is the licensing + order-to-cash + customer-portal system for the whole SmartEPT business. It runs as a multi-tenant SaaS (each customer is a "tenant") and the same codebase also backs on-premises installs through offline node-locked licence files.

The three faces of Central

Super-Admin console (/admin) — Ametecs staff run the business: clients, licences, orders, invoices, pricing, coupons, leads, support, downloads and settings. Access is split across three staff roles (super / sales / support).

Client portal (/client) — each customer signs in to see their licences and invoices, buy or renew, download the agent/server installers, watch their storage, and raise tickets.

Public surfaces — the marketing landing page (data-driven pricing), the checkout pay-link, the gateway webhooks, and the legal pages (privacy / terms / refunds / contact).

2 · Why it is built this way

Central was built to a small set of deliberate principles. Understanding them explains almost every design decision in the rest of this handbook.

One ledger door for money

Every rupee that moves — a Razorpay/Stripe capture, a manual NEFT/UPI/cheque entry, a part-payment on credit, or a refund — passes through a single method, `recordPayment()`. It runs inside a database transaction with a row lock, is idempotent per gateway payment id, provisions the customer on the first money (or credit) event, and settles the order exactly when the ledger covers the total. "Received" and "balance" are always computed from the ledger, never stored, so the books can never silently disagree with the money.

Licences that work offline and are hard to forge

A licence can be delivered as an RSA-2048 signed .lic file that a customer server verifies without ever contacting Central. The private signing key lives only on Central and never ships; the product only carries the public key. Licences are node-locked to a server fingerprint and enforce a seat cap on device activation. This lets Ametecs sell to security-conscious, air-gapped clients while keeping piracy hard.

GST-first invoicing

Because the customer base is Indian businesses, tax correctness is not an afterthought. Invoice numbers run as a consecutive financial-year series (GST style, resets 1 April), the 18% tax is split into CGST+SGST or IGST by comparing buyer and seller state, and SAC code, place of supply, buyer GSTIN and amount-in-words are snapshotted onto every document so a tax record never changes when a profile later does.

Admin-configurable, not hard-coded

Prices, discounts, the setup fee, storage slabs, GST rate, SAC, company and bank details, gateway keys, SMTP and WhatsApp credentials all live in a Settings table and are edited in the console — never in code. The public landing page reads pricing from the API, so a price change in one screen flows to the website within minutes.

Self-service by default, credit by judgement

Customers can sign up, buy, renew and download without a human in the loop; but Ametecs can also raise a quotation, allow a client onto credit (provision immediately, collect the balance later), and chase renewals automatically. Nothing locks a paying client out over the last rupee — credit is a commercial decision the console supports, not a wall.

3 · Architecture at a glance

Central is a standard Laravel application with a thin controller layer over a rich service layer. The controllers validate input and enforce roles; the services hold the commercial logic so the same rules apply whether a request comes from an admin, the portal, a gateway webhook or the daily scheduler.

The service layer (where the logic lives)

Service	Responsibility
PricingService	All commercial arithmetic in one place — Cloud rental by user band + cycle, Perpetual by capacity band, setup fee, AMC, storage slabs, ecosystem discount. Falls back to code constants when a Setting is unset so the money path never breaks on a fresh install.
BillingService	Order → payment → licence → invoice automation. Owns recordPayment() (the ledger door), quotations, renewals (incl. grandfather cap), refunds/credit notes, and FY-series numbering for orders/invoices/credit-notes.
LicenceService	Issues and renews licences, computes expiry from the billing period, builds the signed entitlement bundle a product server caches.
LicenseSigner	RSA-signs the offline .lic token (SHA-256) with the private key that never leaves Central.
RazorpayService / StripeService	Gateway integration — create order/session, verify signatures, read back the authoritative charged amount.
MailService / WaService / OtpService	Queued transactional email (with a mail log), Interakt WhatsApp templates, and OTP issue/verify for portal signup and password reset.
ProductProvisioner	Stands up the hosted admin console for a cloud tenant on first provision (best-effort, never blocks a payment).

Roles & access

Admin staff hold one of three roles — super, sales or support — enforced per route. Read screens are open to all staff; write actions require sales or super; configuration (plans, settings, staff accounts, gateway keys, WhatsApp templates, downloads) is super-only. Customers authenticate on a separate "client" guard that scopes every query to their own tenant, so one customer can never see another's data.

4 • The data model

The tables below are the backbone. Every customer-owned row carries a `tenant_id`; every money document is immutable once issued.

Table	What it holds
tenants	One customer company — profile, GSTIN, state, currency, deployment (cloud/client-hosted), trial dates, purge window, storage quota + alert level, console URL.
tenant_users	The login accounts inside a customer (portal users).
licences	The entitlement: key, plan, kind (trial / subscription / perpetual), billing period, deployment, device (seat) limit, features, status, starts/expires, grace days, AMC expiry, server fingerprint. NEW: pricing_model (legacy/v2) + legacy_baseline_inr for grandfathering.
licence_devices	Each activated endpoint under a licence — enforces the seat cap and node-lock.
orders	A deal: number, optional quote_number, requested_by, NEW po_number + valid_until, line items, subtotal/tax/total, gateway, status (quote/created/paid/failed/refunded), credit due date.
order_payments	The money ledger — one row per capture / instalment / refund (negative row + credit-note number). received() and balance() sum this.
invoices	The GST tax invoice — FY-series number, CGST/SGST/IGST split, SAC, place of supply, buyer GSTIN, amount, status.
plans + plan_volume_tiers + plan_perpetual_bands	The single SmartEPT plan, its Cloud per-user volume tiers, and its Perpetual by-capacity bands.
coupons	Percent/fixed discount codes with validity window, min users, exclusivity and usage caps.
leads	Sales pipeline captured from the landing page and demos.
support_tickets + support_ticket_messages	Client tickets with an append-only threaded timeline (author + status changes).
download_artifacts + download_logs	The managed installer catalogue (agent Win/Mac/Linux + server) and per-client download quota history.
settings	Every admin-editable knob — pricing, GST, company/bank, gateway keys, SMTP, WhatsApp, and the locked numbering counters.
audit_logs	Append-only record of every meaningful action, indexed by company + time, exportable to CSV.
webhook_events	De-duplication + error log for gateway webhooks.
storage_usage	Daily GB-used per cloud tenant, for metered storage billing and the 90%/100% governance.

5 • The order-to-cash flow

This is the heart of Central. A deal moves through the same pipeline whether it starts from an admin, the portal or a self-service signup.

- **Quote / order.** An admin or a client raises a deal. As a quotation it gets a quote number and prints as a "Quotation / Proforma Invoice"; as an order it is payable immediately. An optional client PO number rides along, and the pay-link is stamped with a validity date (default 7 days).
- **Pay-link.** Every order exposes `/pay/{number}/{token}`, where the token is an HMAC of the order number — safe to send on WhatsApp or email. The page shows total, received-so-far and balance, and collects the balance via Razorpay (UPI/card/netbanking) or Stripe (international card). An expired quotation refuses payment with a friendly message.
- **Payment recorded.** The capture (or a manual NEFT/UPI/cheque entry, or a part-payment on credit) lands in `recordPayment()`. It writes one ledger row, provisions on the first event, and settles when the balance hits zero.
- **Provisioning.** On that first event the licence is issued or renewed, the tenant becomes active, the setup fee is marked consumed if it was on the order, any coupon is redeemed once, and the GST tax invoice is cut — marked paid if the ledger already covers the total, otherwise issued as "due" with the credit due date.
- **Receipt.** When the balance reaches zero the customer gets a full receipt email (and a WhatsApp confirmation if configured); part-payments get a warm acknowledgement showing the live balance.

Credit, partial payments and refunds

Partial / due: an admin can provision a client on part-payment or full credit; the workspace goes live immediately and the balance is chased later on the same pay-link — nothing locks over the last rupee.

Refund: a refund is a negative row in the same ledger with its own GST credit-note number, so `received()` drops automatically and the books stay consistent. It can never exceed what was received.

6 • Pricing model v2

SmartEPT is sold as one all-features product, priced purely by the number of users. There are no plan tiers to choose between.

Element	How it works
SmartEPT Cloud (rental)	Per user / month, chosen by user band. Advance-period discounts kept: Annual is the published rate, Half-Yearly is ~10% above it, Quarterly is the base. Hosting and storage are included — no separate storage line.
SmartEPT Perpetual	One-time, client-hosted, all features, priced by user-capacity band (e.g. 30 users, 100, 250 ...). Above the top band it becomes a custom quotation.
Setup & Onboarding	Always extra, self-service: ₹5,000 covers 30 users + ₹100 per extra user (uncapped, per the 27-Jul decision).
AMC (Perpetual, optional)	~18% of the prevailing band price per year; year-1 updates/support are included, AMC begins from year 2.
Ecosystem discount	A flat 10% for existing SmartDCM / SmartPRS customers, on both models.
Grandfather	Existing pre-v2 customers migrate to the user-based model at their next renewal, with a price cap (default 20%, admin-editable) so the bill

	cannot jump sharply; after that they are on v2.
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Every number above is admin-editable in Plans and Settings; the landing page follows within about five minutes.

7 • The licensing engine

A licence is the key that unlocks the product for a customer. Central can enforce it two ways, and both are built.

- **Online entitlement.** A product server phones home and caches a signed entitlement bundle (plan, seats, features, expiry) — HMAC-signed over canonical JSON.
- **Offline .lic file.** For air-gapped or on-prem clients, Central issues an RSA-2048 signed token the server verifies locally with the public key. It is node-locked to the first server fingerprint that presents it, carries an HMAC check digit, and respects the licence's own grace period.
- **Seat cap.** Each device activation consumes a seat; the cap is enforced at activation. An admin can move a licence to a new server via a fingerprint reset.
- **Lifecycle.** Trials block the instant they end (zero grace); paid licences carry a grace period; perpetual licences never expire but gate updates/support behind AMC.

8 • GST & compliance

Central issues documents that stand up to an Indian audit.

- Invoice, quote and credit-note numbers each run as a consecutive financial-year series (PREFIX-FY-MM-count), generated as MAX+1 under a row lock so a deletion or a restored backup can never duplicate a number.
- The 18% GST is split as CGST 9% + SGST 9% for buyers in the seller's state (Telangana / 36) or IGST 18% for other states; the split is a breakdown of the same total, never a change to it.
- SAC code (997331 software licensing), place of supply, buyer GSTIN and amount-in-words are snapshotted onto the document at issue time.
- USD invoices carry 0% GST (export of services). Public legal pages — privacy, terms, refunds, contact — are live and linked, as gateways require.

9 • Lifecycle & automation

A daily scheduled command keeps the business running without manual chasing. It depends on the operating system running php artisan schedule:run every minute — this is the single most important operational check.

Automation	What it does
Renewal dunning	Reminds a customer before a subscription licence expires at T-30, T-7, T-1 and T-0, over email (and WhatsApp when configured).
Trial reminders	Nudges a trial at T-3, T-1, T-0 and marks it expired; a purge window later removes data while keeping billing/GST records.
Storage governance	Warns a cloud tenant at 90% of quota and pauses screenshots at 100%.
Lapsed-licence expiry	Expires licences past their grace and closes out the lifecycle.

(Product side) archives & attendance

On the SmartEPT product, the same scheduler builds employee-archive ZIPs and runs nightly attendance/alerts — so the check matters there too.

Known gap (from the 27-Jul deep-scan)

AMC expiry has no reminder yet. With the v2 Perpetual model this matters — a perpetual client can silently drop out of updates/support. Adding AMC dunning (T-30/7/1/0 on `amc_expires_at`) is a tracked quick win.

10 · Security posture

The 27-Jul code review found the foundations sound and flagged hardening to do before the client base scales.

What is solid

- Tenant isolation: every portal query and every document print checks tenant ownership — no cross-customer access (no IDOR).
- Gateway webhooks verify Razorpay/Stripe signatures with constant-time comparison, fail closed when a secret is unset, and de-duplicate by event id.
- Passwords are hashed; auth and public endpoints are rate-limited; admin roles are enforced per route.
- The licence private key never ships; offline tokens are RSA-signed and node-locked.

Hardening still to do

- **P1 — Encrypt gateway/mail secrets at rest.** Today the Settings table stores gateway keys and the mail password in plaintext (masked in the UI only). Encrypt them, or move to env / a secrets manager.
- **P1 — Admin 2FA.** The console is password-only; add TOTP 2FA at least for the super role — it guards the whole money console.
- **P2/P3.** Shorten the admin session window; add a unique index on `webhook_events(gateway,event_id)`; use the licence's real grace in the .lic payload; handle failed-payment webhooks.

11 · The Super-Admin console, screen by screen

The console lives at /admin and opens on the Dashboard. Menus are grouped by lifecycle. Each screen below lists what it does and how, why it matters, and which staff roles can use it (S = Super, Sa = Sales, Su = Support).

Screen	What it does & how	Why it matters	Who
Dashboard	Live counts and money KPIs — active clients, trials, licences, revenue, outstanding — as the landing view.	A one-glance pulse of the business every time staff log in.	All
Clients (Tenants)	Create and edit customer companies; capture GSTIN, state, currency, deployment; drill into a client's licences, orders and storage.	The master customer record every document and licence hangs off.	S · Sa
Licences	Issue directly, renew, suspend/resume, edit seat limit, generate an offline .lic file, reset a server fingerprint, deactivate a device.	The entitlement control panel — what each customer is actually allowed to run.	S · Sa
Orders & Quotes	Raise an order or quotation (now with a PO number and 7-day validity), approve a quote, mark paid / partial / due, record a later balance, refund with a credit note, copy the pay-link.	The deal desk — where money is committed and collected.	S · Sa
Credit clients	Every order provisioned before full payment, with total / received / balance / due date; overdue turns red. Follow-up is manual by design.	Keeps credit exposure visible without ever auto-locking a paying client.	All (write S·Sa)
Invoices	Browse and print GST tax invoices; each carries the CGST/SGST/IGST split, SAC and buyer GSTIN.	The audit-ready tax record.	All
Trials	Active and expired trials; extend a trial by a number of days.	Converts free trials into paid customers before they lapse.	All (write S·Sa)
Storage	Per-cloud-tenant GB used and the monthly storage charge for a chosen month.	Meters cloud hosting cost and flags heavy users.	All
Plans & Pricing	Edit the single SmartEPT plan — annual rate (Quarterly/Half-Yearly derive automatically), storage, USD, min users; view the by-user Perpetual bands. Legacy per-device columns were removed.	The one place prices are set; the landing page follows within ~5 minutes.	S
Coupons	Create percent/fixed discount codes with validity, min-users, exclusivity and usage caps.	Runs campaigns and per-deal discounts without touching base prices.	S · Sa
Leads	The sales pipeline captured from the landing page and demos.	Nothing from the website falls through the cracks.	S · Sa
Support desk	Threaded tickets with an append-only timeline and IST timestamps; reply	A real support history that can never overwrite a prior	All (write S·Sa)

	(emails the client) and change status.	reply.	
Users & Roles	Manage super/sales/support staff logins, reset passwords, deactivate; a read-only permission matrix. Guardrails keep at least one active super and prevent self-lockout.	Least-privilege staff access to a money console.	S
Downloads	Upload/publish the agent (Win/Mac/Linux) and server installers; set per-client download quotas and view the history.	One managed place clients pull the right installer from.	S
WhatsApp templates	Manage and test Interakt template messages used for payment and reminder notifications.	Keeps the WhatsApp channel compliant and working.	S
Reports (Accountant)	GST register, Collections and Outstanding as CSV downloads.	Hands the accountant clean, ready figures.	S · Sa
Audit log	Filter the append-only action log by date-range, subject and IP; export to CSV (secrets redacted).	Accountability and compliance evidence.	All (export gated)
Settings	Company & bank details, GST rate and SAC, pricing knobs (incl. grandfather cap % and quote validity days), gateway keys, SMTP and WhatsApp credentials, timezone.	The single configuration surface — nothing here is hard-coded.	S
Help → Troubleshooting	Live System Health self-check (scheduler, product link, licence key, biometric, screenshots) + a plain-language Known-Issues KB + a log viewer.	Non-technical self-service diagnosis at a client site.	S

12 · The client self-service portal

The portal lives at /client. A customer signs up with an OTP-verified email, or logs in; every screen is scoped to their own tenant.

Screen	What it does & how	Why it matters	Who
Login / Signup	Email + password login; self-signup with an OTP to the email; forgot-password via OTP. Rate-limited.	Lets customers onboard themselves without Ametecs staff.	Client
Overview	A dashboard of the customer's licences, recent orders, invoices and storage at a glance.	The customer's home base.	Client
Licences	Their active licences with seats, expiry and status; one-click renew.	Self-service renewal reduces churn and support load.	Client
Buy / Quote	Estimate and buy on the user-based model (Cloud rental or Perpetual, now with an optional PO number), or raise a quotation for management to pay.	Turns the portal into a sales channel.	Client
Orders & Invoices	Their orders with pay-links, and printable GST invoices / the Quotation-Proforma print.	Everything they need for their own accounts team.	Client
Downloads	The published agent/server installers for their OS, within their download quota.	The correct installer, always current, quota-controlled.	Client
Storage	Their cloud storage usage against quota, with the 90%/100% governance visible.	No surprise when screenshots pause at 100%.	Client
Support	Raise and track tickets against the same threaded timeline staff see.	A real two-way support channel.	Client
Account	Edit the billing profile (GSTIN, state, address), change password.	Keeps their tax details correct on every future invoice.	Client
Open Console (SSO)	One-click single sign-on into their hosted SmartEPT admin console (cloud tenants).	No second password to reach the product itself.	Client

13 · Checkout & payments

The public pay page at /pay/{number}/{token} is the one place money is collected online. The token is an HMAC of the order number, so the link is safe to forward. The page shows total, received and balance, and — for a credit client — stays alive to collect the remaining balance. Razorpay handles UPI, cards and netbanking for INR; Stripe handles international cards. A quotation that has passed its validity date refuses payment and points the customer to sales. On success the ledger records the exact charged amount read back from the gateway, provisions the customer, and emails the receipt.

14 · Deployment & operations

Central runs on Laragon (Apache + PHP 8.3 + MySQL) in production at Ametecs and installs the same way at an on-prem client. The operational essentials:

- **The scheduler is critical.** Windows Task Scheduler (or cron) must run `php artisan schedule:run` every minute — it drives dunning, trial reminders, storage governance and, on the product side, archive builds and attendance. The Troubleshooting screen turns this into a green/red tile.
- **Uploads & PHP.** Raise `upload_max_filesize` / `post_max_size` to 256M for installer uploads, and keep the PHP zip extension on for archive ZIPs; restart PHP after a change.
- **Secrets live in Settings.** Gateway keys, SMTP and WhatsApp credentials are entered in the console, not in code — see the security note about encrypting them at rest.
- **Licensing.** The RSA private key stays on Central; on-prem clients receive a node-locked `.lic` file. SaaS and on-prem share the same architecture.
- **Migrations after every update.** Run `migrate.bat` after pulling changes; the latest is `2026_07_27_000200` (pricing-finalise: PO number, quote validity, grandfather fields, settings).

15 · Where Central is going

Central is a mature, well-architected platform already running the order-to-cash and licensing flow correctly. The forward work, in priority order, is: (1) the two P1 security items — encrypt secrets at rest and admin 2FA; (2) AMC-expiry dunning to protect perpetual recurring revenue; (3) small correctness/robustness fixes (real grace in the `.lic` payload, a unique webhook index, a shorter admin session, failed-payment handling); then product depth — self-serve storage top-up, optional card-on-file auto-renewal, GST e-invoicing (IRN + signed QR) as turnover grows, and a data-subject request path. None of these block running the business today.

Appendix A · Route map

Route group	Endpoints
Public	/ (landing) · /privacy /terms /refunds /contact · /pay/{number}/{token} (+ razorpay/stripe) · /webhooks/razorpay /webhooks/stripe
Client auth	/client/login · /client/signup (request-otp, verify) · /client/forgot (request-otp, reset) · /client/logout
Client portal (client.auth)	/client · /client/console (SSO) · invoices/{id}/print · orders/{id}/quote-print · download/{artifact} · api: overview, downloads, licences, orders, invoices, plans, storage, quote, orders(create), licences/{id}/renew, account/billing, account/password, tickets
Admin auth	/admin/login · /admin/logout
Admin console (admin.auth)	/admin · invoices/{id}/print · orders/{id}/quote-print · api reads: dashboard, tenants, licences, orders, credit-clients, invoices, trials, storage, plans, webhooks, audit, leads, coupons, tickets, quote
Admin write (super,sales)	tenants, licences (+action, license-file, limit, deactivate-device), orders (create, mark-paid, record-balance, approve-quote, setup-invoice), trials/extend, storage, leads, coupons, tickets, reports (gst-register, collections, outstanding)
Admin super-only	plans/{plan}, settings (get/put), config/test-email, wa-templates, diagnostics, logs, download-artifacts (+limits), admin-users (+reset-password)

Appendix B · Roles & permissions

Role	Scope	Can do
Super	Everything	All of Sales + configuration: plans & pricing, settings, gateway keys, WhatsApp templates, downloads catalogue, staff accounts & roles, diagnostics, audit export.
Sales	Deals & customers	Create/edit clients, issue/renew/edit licences, raise & collect orders, refunds, trials, leads, coupons, reply to tickets, accountant reports.
Support	Read + tickets	Read all screens; work the support desk. No money or configuration writes.
Client	Own tenant only	Portal: buy, renew, download, view invoices, manage storage, raise tickets, edit own billing profile, SSO to console.

Appendix C · Key settings

All editable in Settings (super role). A missing value falls back to a safe code default.

Setting	Meaning
pricing_annual_discount_pct / half_yearly_discount_pct	Advance-cycle discounts the Quarterly/Half-Yearly rates derive from.

pricing_setup_base_inr / included_devices / per_extra_inr	Setup fee: ₹5,000 / 30 users / ₹100 per extra (uncapped).
pricing_grandfather_cap_pct	Max % a grandfathered renewal may rise over the previous rate (default 20).
quote_validity_days	How long a quotation / pay-link stays valid (default 7).
gst_rate / sac_code / seller_state_code	Tax rate, SAC and seller state for the GST split.
company_* / bank_* / upi_id	Letterhead + bank/UPI block on invoices and quotes.
razorpay_* / stripe_* / mail_* / interakt_api_key	Gateway, email and WhatsApp credentials (flagged: encrypt at rest).
display_timezone	Console display timezone (default Asia/Kolkata / IST).

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